

TASK:

- ✓ [A] Take 10 patients
- ✓ [B] Select the first and the last time point you have (however far they are apart)
- ✓ [C] Load them into Matlab
 - Normalize the intensities to [0,1]
 - Compute $1 - \text{abs}(m1 - m0)$, where $m1$ is the last time point and $m0$ is the first time point
 - Store a picture for the axial, coronal, and sagittal view
- ✓ [D] Perform an affine registration using NIPAV:
 - Load two images
 - Algorithms - Registration - Optimized Automatic Registration
 - Register time point 0 (first time point) to time point 1 (last time point)
 - Use the default settings (Affine registration, 12 degrees of freedom; trilinear interpolation; correlation ratio as similarity and Powell's method for optimization)
 - This will create a new image, called "originalname_register"
 - Store image
- ✓ [E] Repeat [C], now for the reference image (image $m1$ at time point 1) and the registered image you created under [D]
- ✓ [F] Write a report about the results (i.e., explain what you did and add the images you created as figures)

Report:

- done for 11 patients in the folder: '/project/mang/data/ADN_data analysis' on Sabine

Description & notations:

File Name

Decks

patient ID - orig - similarity .mat

similarity measure between original image at first time point & last time point

patient ID - orig - similarity A. png

Axial view of the image similarity

" S. png

Sagittal view "

" C. png

Coronal view "

patient ID - registered to 1 .nii

registered image at first time point to image at last time point using NIPAV

patient ID - reg - similarity .mat

similarity measure between registered image at first time point & original image at last time point

patient ID - reg - similarity A. png

Axial view of the image similarity

" S. png

Sagittal view "

" C. png

Coronal view "

[C] - Load & normalize image intensities in MATLAB:

Code: nifti_normalize.m

using the following formula where $\text{newMin} = 0$ & $\text{newMax} = 1$

From Wikipedia, the free encyclopedia

In image processing, **normalization** is a process that changes the range of *pixel* intensity values. Applications include photographs with poor contrast due to glare, for example. Normalization is sometimes called contrast stretching or *histogram stretching*. In more general fields of data processing, such as *digital signal processing*, it is related to as *dynamic range expansion*.^[1]

The purpose of dynamic range expansion in the various applications is usually to bring the image, or other type of signal, into a range that is more familiar or normal to the senses, hence the term normalization. Often, the motivation is to achieve consistency in dynamic range for a set of data, signals, or images to avoid mental distraction or fatigue. For example, a newspaper will strive to make all of the images in an issue share a similar range of grayscale.

Normalization transforms an *n*-dimensional *grayscale* image $f : \{X \subseteq \mathbb{R}^n\} \rightarrow \{Min, \dots, Max\}$ with intensity values in the range $[Min, Max]$, into a new image $I_N : \{X \subseteq \mathbb{R}^n\} \rightarrow [newMin, \dots, newMax]$ with intensity values in the range $[newMin, newMax]$.

The *linear* normalization of a *grayscale* digital image is performed according to the formula

$$I_N = \frac{(I - Min) * (newMax - newMin)}{Max - Min} + newMin$$

For example, if the intensity range of the image is 50 to 180 and the desired range is 0 to 255 the process entails subtracting 50 from each of pixel intensity, making the range 0 to 130. Then each pixel intensity is multiplied by 255/130, making the range 0 to 255.

Normalization might also be non linear, this happens when there isn't a *linear* relationship between I and I_N . An example of non-linear normalization is when the normalization follows a *sigmoid function*, in that case, the normalized image is computed according to the formula

$$I_N = (newMax - newMin) \frac{1}{1 + e^{-\frac{I - \beta}{\alpha}}} + newMin$$

Where α defines the width of the input intensity range, and β defines the intensity around which the range is centered.^[2]

Auto-normalization in image processing software typically normalizes to the full dynamic range of the number system specified in the image file format.

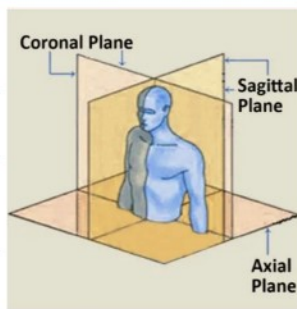
- Computing $1 - |m1 - m0|$ & visualizing in MATLAB:

Code: `niftiImage_similarity.m`

$|m1 - m0| \rightarrow$ dissimilarity

$\therefore 1 - |m1 - m0| \rightarrow$ similarity

- storing axial, sagittal & coronal view of the image



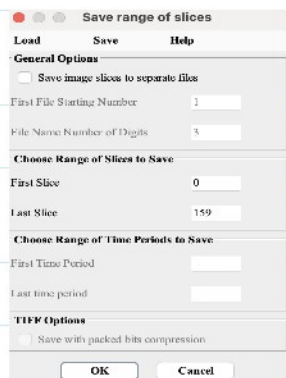
[D] Registration using HIFAV

- First open the image at last time point

- then " " " " first " "

- registered image at first time point to last time point with default setting

- saved as a nifti file with the following setting:



[E] Repeat [C] using registered image at first time point & the original image at last time point

Results & Observations: Similarity Measure ($1 - |m_1 - m_0|$)

$m_0 \rightarrow$ original image at first time point

$m_1 \rightarrow$ " " " last " "

$\tilde{m}_0 \rightarrow$ registered " " first " "

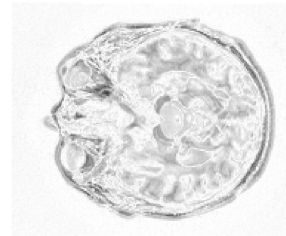
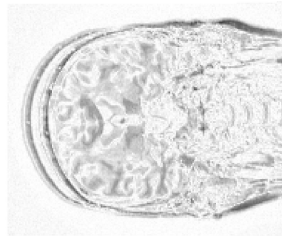
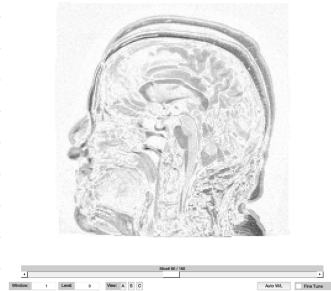
Patient ID: 141-S-1004

views $\rightarrow$ Axial

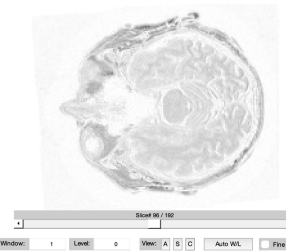
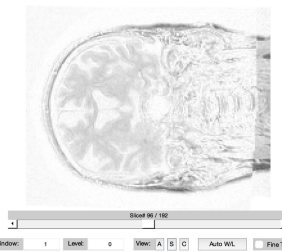
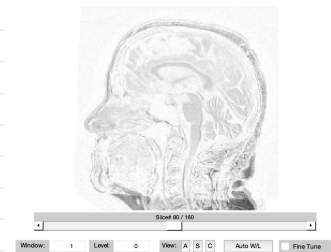
Sagittal

Coronal

type
 $\downarrow$



$1 - |m_1 - m_0|$



$1 - |m_1 - \tilde{m}_0|$

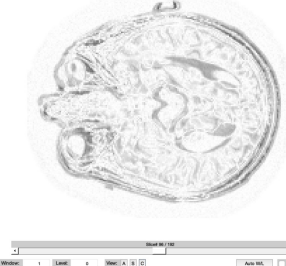
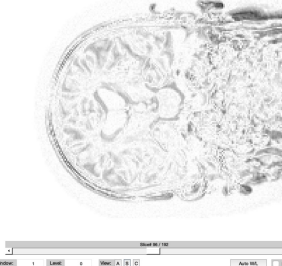
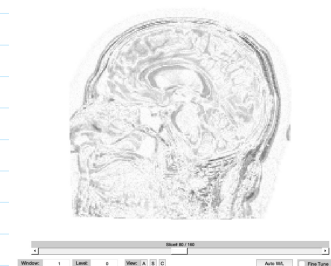
Patient ID: 141-S-1052

views $\rightarrow$ Axial

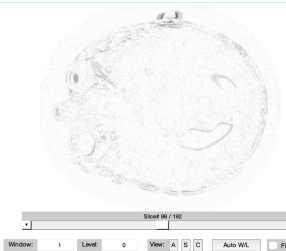
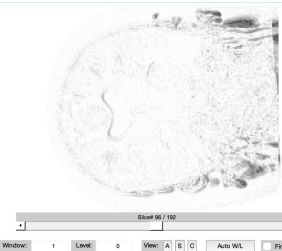
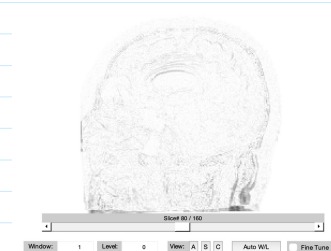
Sagittal

Coronal

type
 $\downarrow$



$1 - |m_1 - m_0|$



$1 - |m_1 - \tilde{m}_0|$

Patient ID: 141-S-1094

views $\rightarrow$ Axial

Sagittal

Coronal

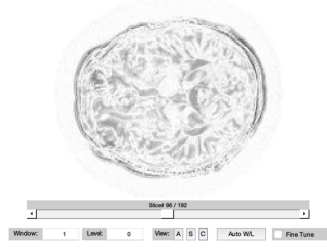
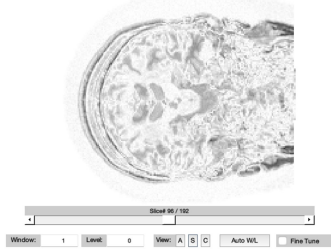
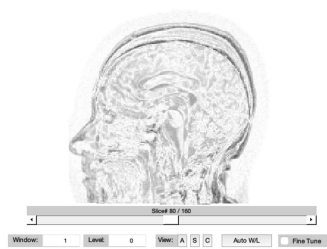
type
 $\downarrow$

views → Axial

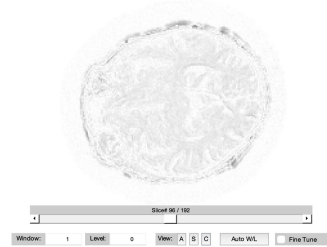
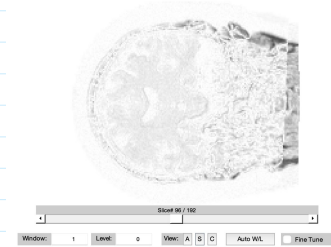
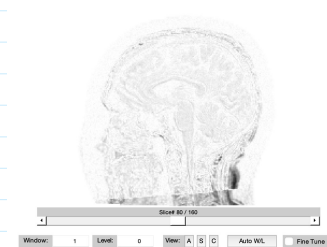
Sagittal

Coronal

type
↓



$$1 - |m_1 - m_0|$$



$$1 - |m_1 - \tilde{m}_0|$$

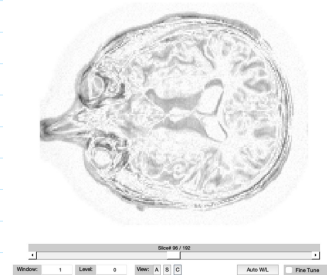
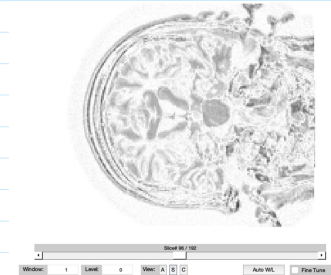
Patient ID: 141-S-1137

views → Axial

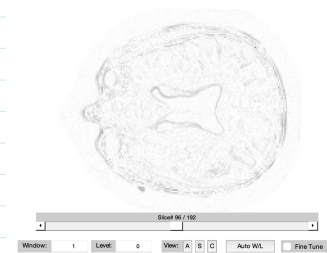
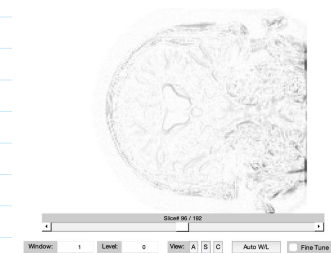
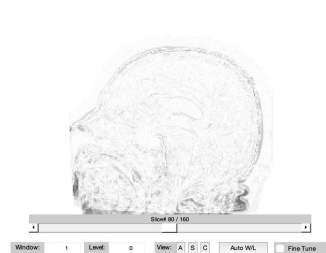
Sagittal

Coronal

type
↓



$$1 - |m_1 - m_0|$$



$$1 - |m_1 - \tilde{m}_0|$$

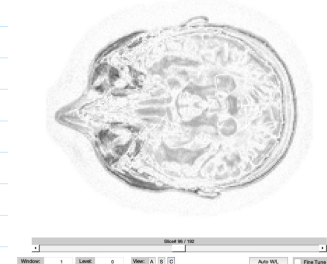
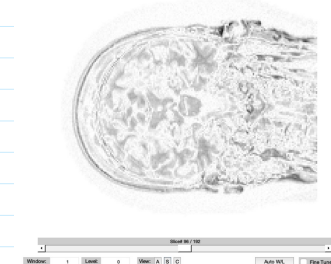
Patient ID: 141-S-1152

views → Axial

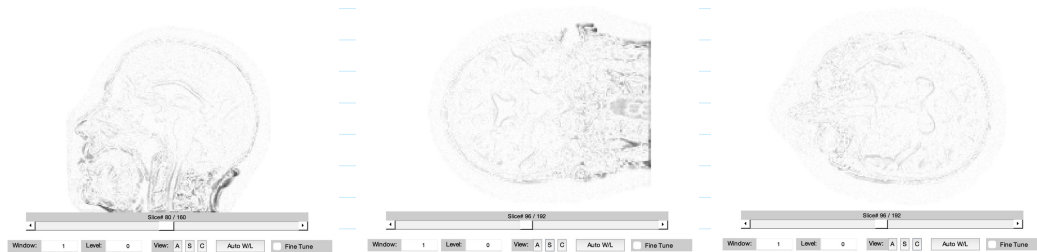
Sagittal

Coronal

type
↓



$$1 - |m_1 - m_0|$$



$$1 - |m_1 - \tilde{m}_0|$$

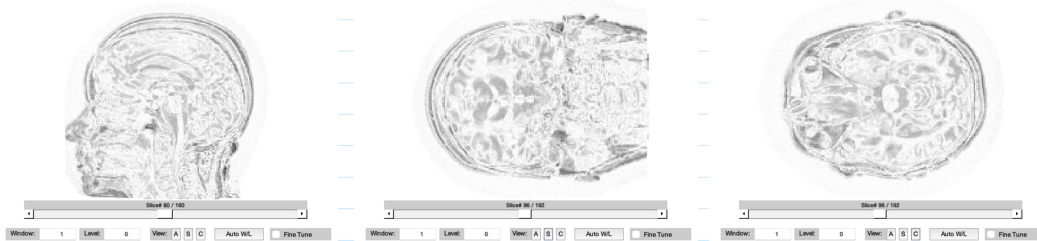
Patient ID: 141-S-1245

views → Axial

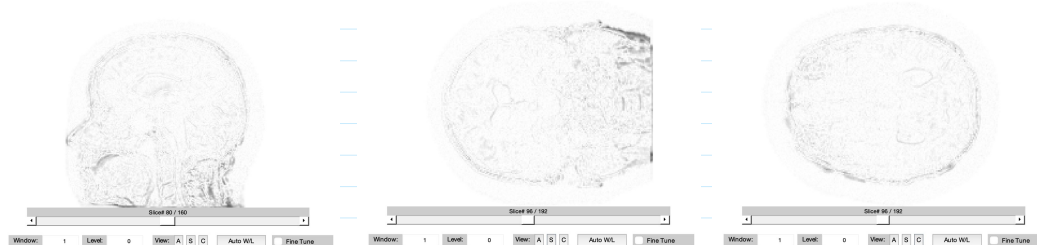
Sagittal

Coronal

type
↓



$$1 - |m_1 - m_0|$$



$$1 - |m_1 - \tilde{m}_0|$$

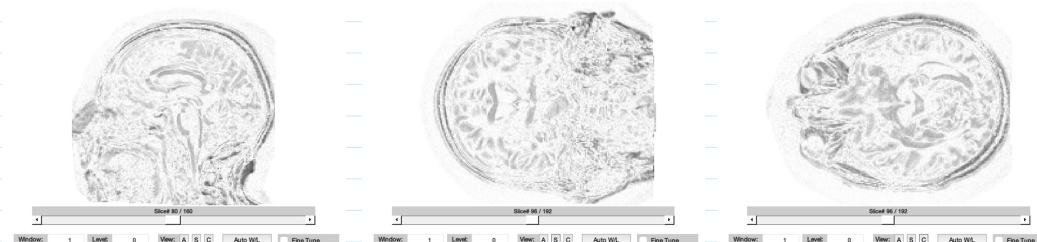
Patient ID: 141-S-1255

views → Axial

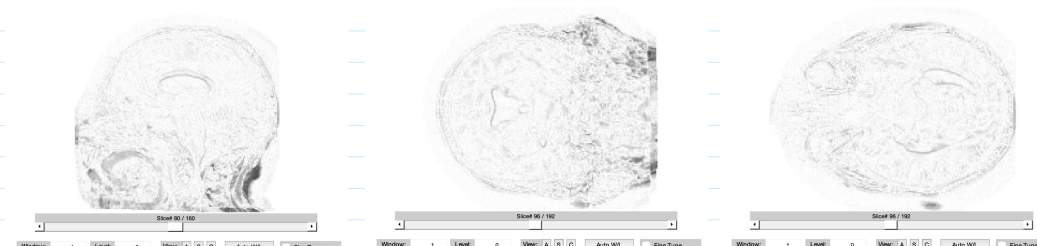
Sagittal

Coronal

type
↓



$$1 - |m_1 - m_0|$$



$$1 - |m_1 - \tilde{m}_0|$$

Patient ID: 941-S-1194

views → Axial

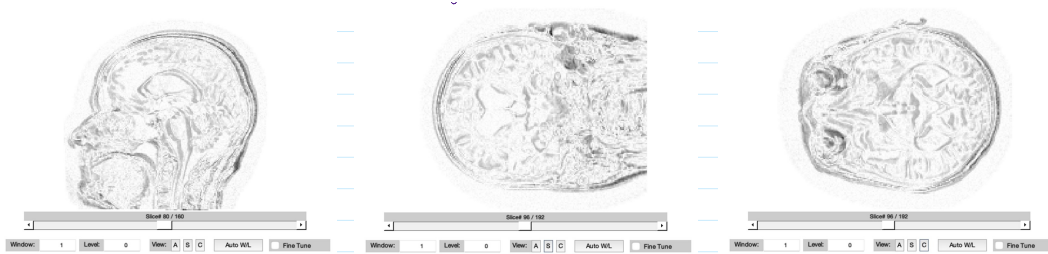
Sagittal

Coronal

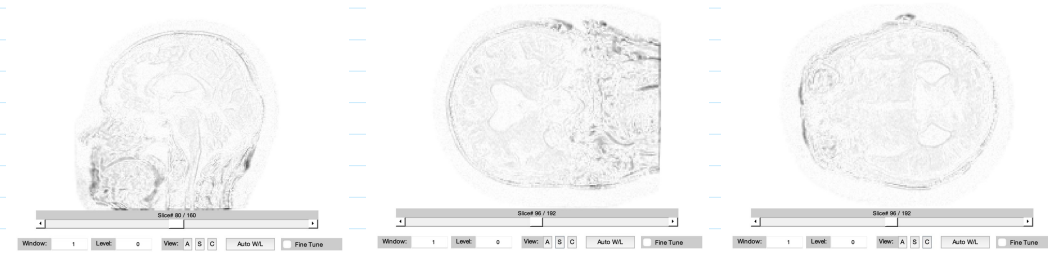
type
↓



$$1 - |m_1 - \tilde{m}_0|$$



$$1 - |m_1 - m_0|$$



$$1 - |m_1 - \tilde{m}_0|$$

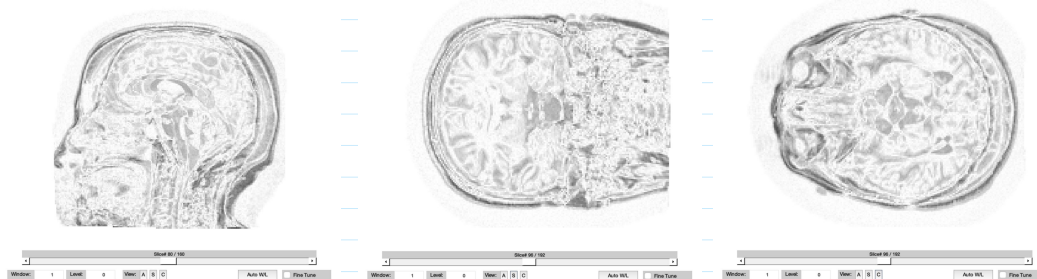
Patient ID: 941-S-1197

views → Axial

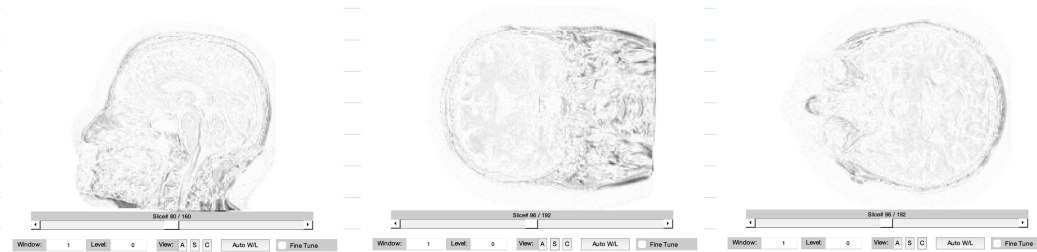
Sagittal

Coronal

type
↓



$$1 - |m_1 - m_0|$$



$$1 - |m_1 - \tilde{m}_0|$$

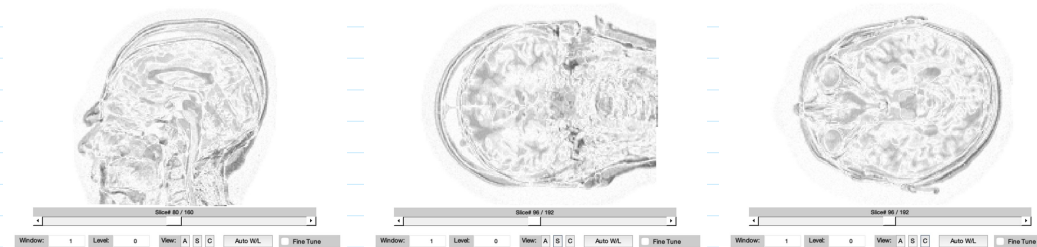
Patient ID: 941-S-1202

views → Axial

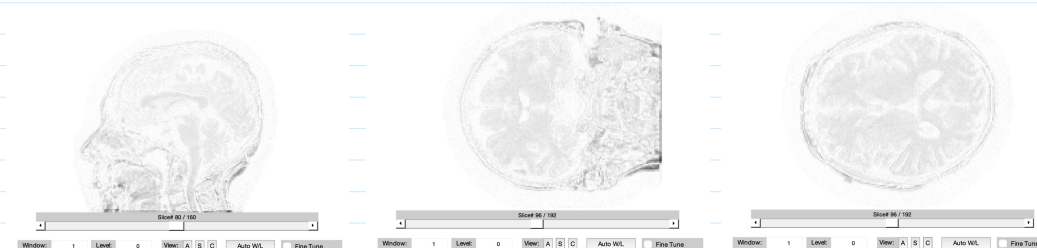
Sagittal

Coronal

type
↓



$$1 - |m_1 - m_0|$$



$$1 - |m_1 - \tilde{m}_0|$$

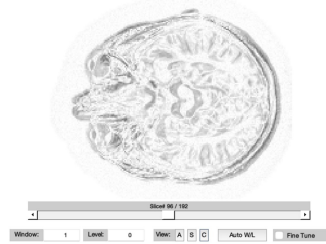
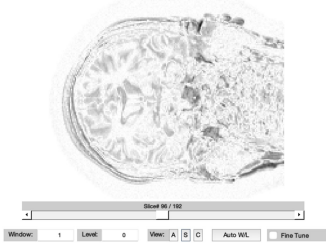
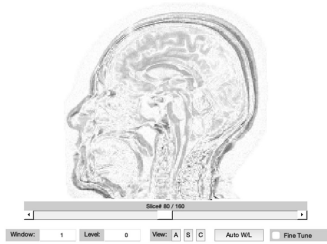
Patient ID: 941-S-1311

views → Axial

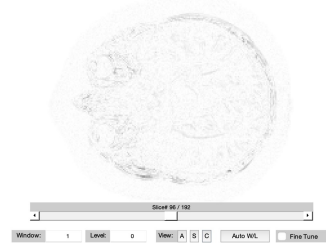
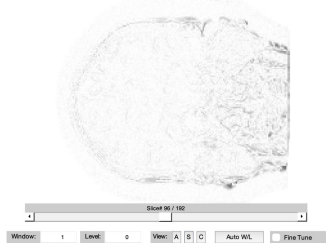
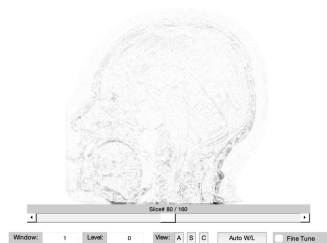
Sagittal

Coronal

type
↓



$$1 - |m_1 - m_0|$$



$$1 - |m_1 - \tilde{m}_0|$$